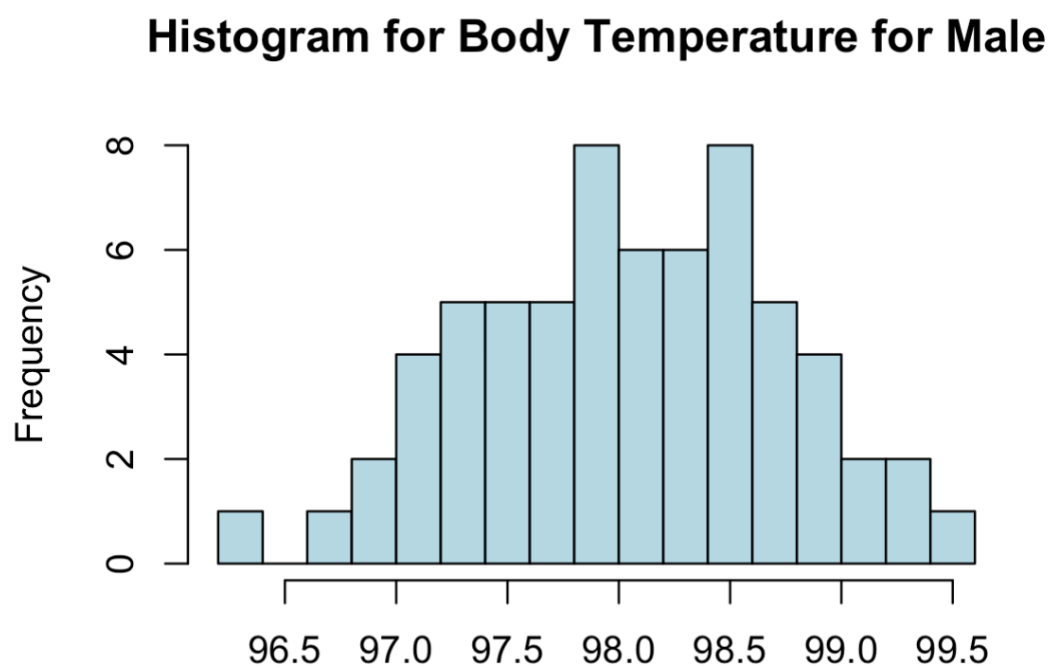
Statistical Methods for Data Science (Spring 2026)
Mini Project #4
Name : FATHIMA FARIDA MOHAMMAD

SECTION 1

QUESTION 1

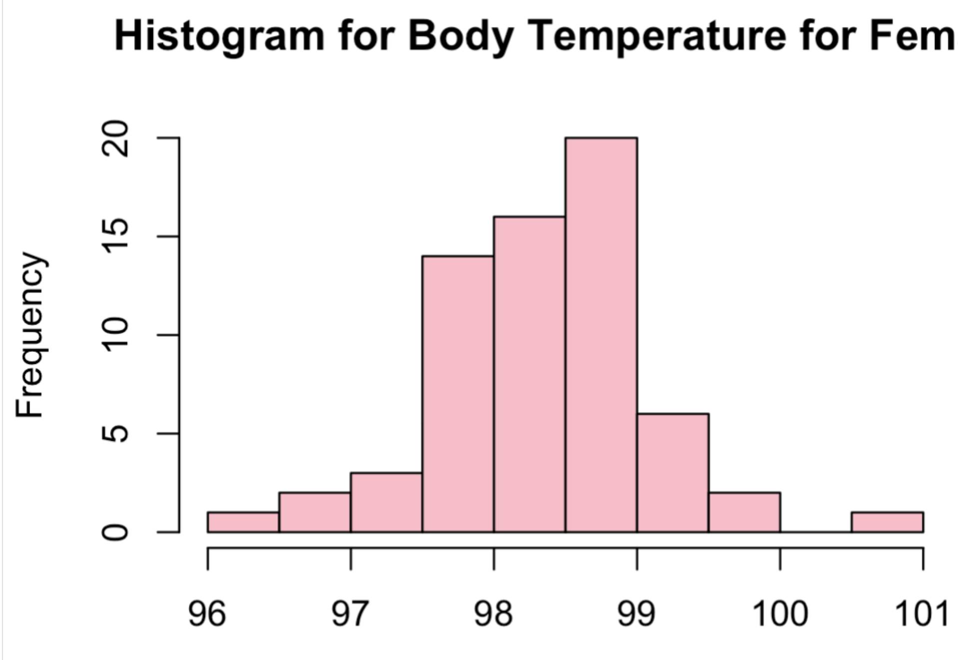
a. Exploratory Data Analysis for Body Temperature by Gender



- Body temperature data for male appears to be approximately normal and symmetric centered around a mean and median of 98.1 F.

- Summary and Statistics for Male Body Temperature:

```
> ## Summary statistics for Male
> summary(filter(heartrate, gender == 1)$body_temperature)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   96.3    97.6    98.1    98.1    98.6    99.5
> sd(filter(heartrate, gender == 1)$body_temperature)
[1] 0.6987558
> range(filter(heartrate, gender == 1)$body_temperature)
[1] 96.3 99.5
```

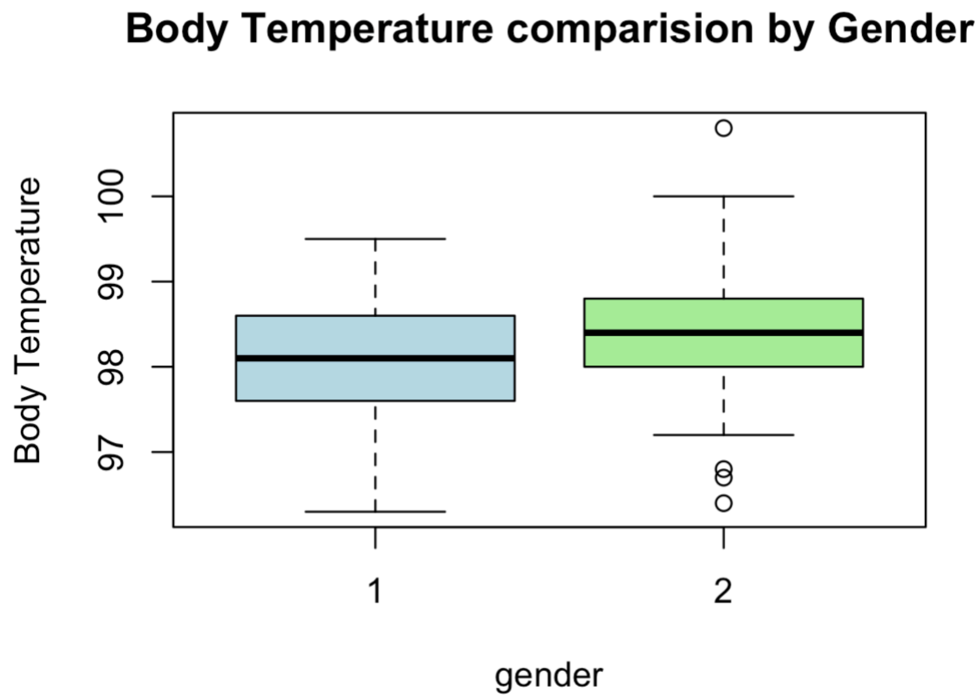


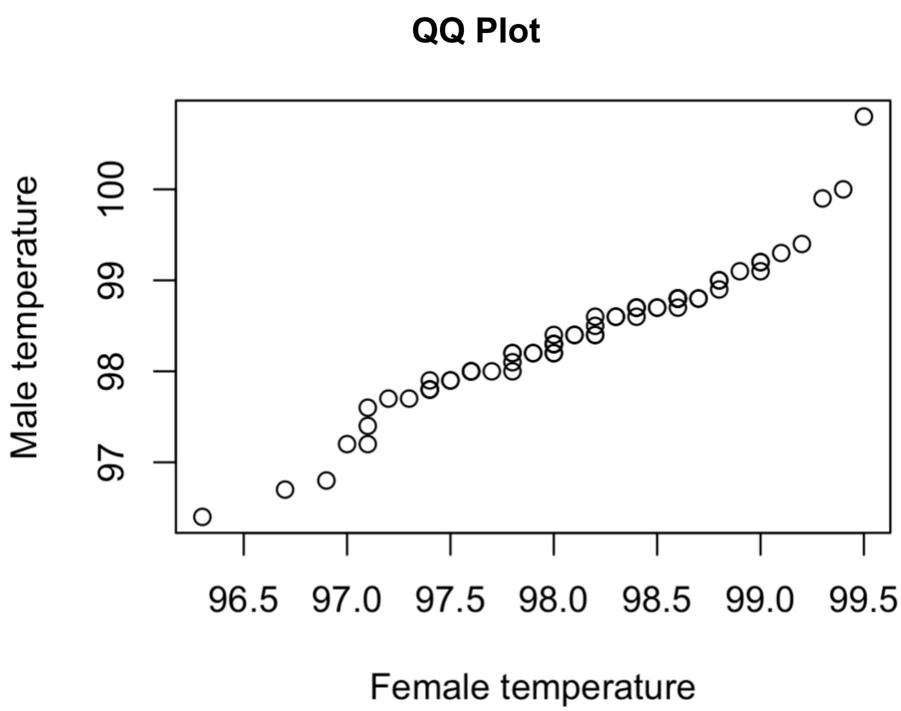
- The female body temperature distribution appears approximately normal and slightly left-skewed.

- Summary and Statistics for Female Body Temperature:

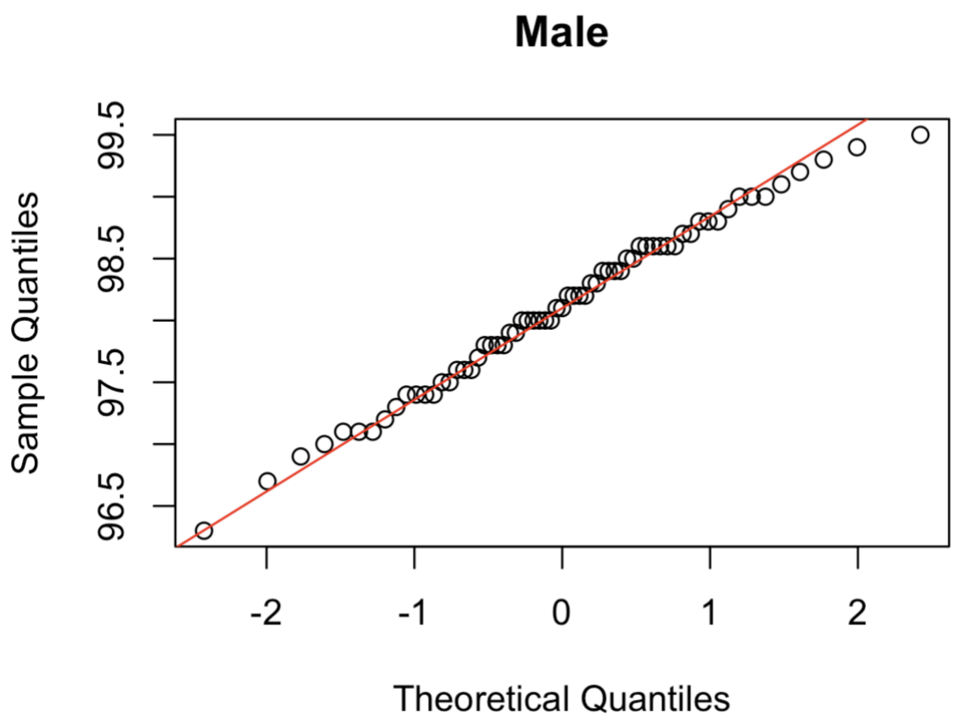
```
> ## Summary statistics for Female
> summary(filter(heartrate, gender == 2)$body_temperature)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  96.40   98.00   98.40   98.39   98.80  100.80 
> sd(filter(heartrate, gender == 2)$body_temperature)
[1] 0.7434878
> range(filter(heartrate, gender == 2)$body_temperature)
[1]  96.4 100.8
```

- The side-by-side boxplot shows that the distribution of body temperature readings differs between the two genders. Females have a higher median body temperature compared to Males, indicating that on average, temperature readings are higher for females. Also, the female group exhibits a wider overall range and more prominent outliers.

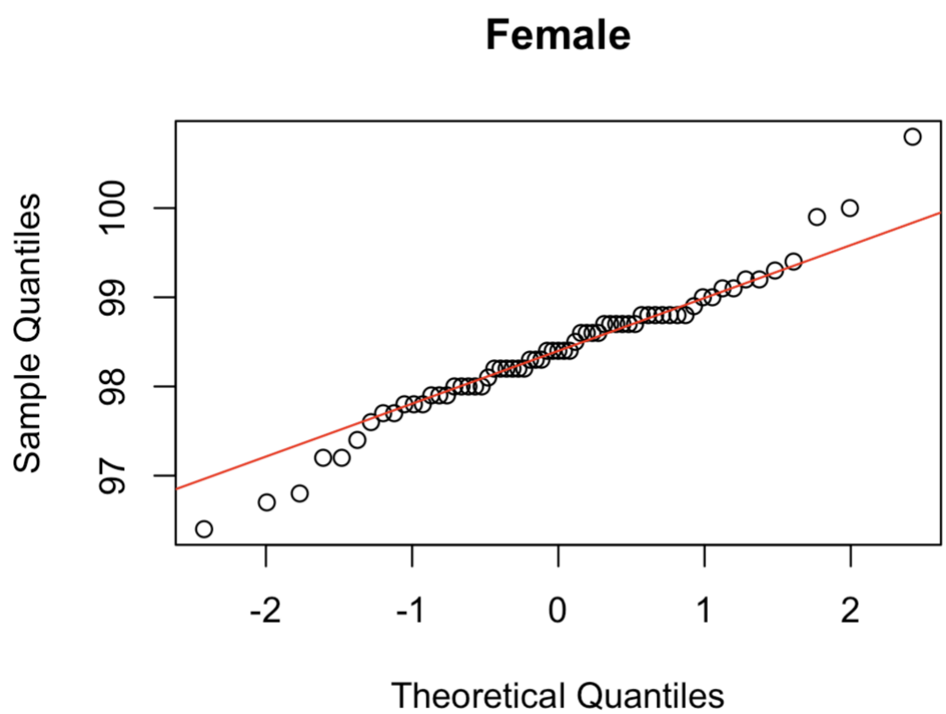




- The QQ plot indicates that even though the two genders follow a nearly identical pattern through the middle of the distribution, the female data exhibits more prominent outliers at both the extremes.



- The male plot shows a very strong linear pattern, indicating a high degree of normality across the entire range.



- The female plot exhibits more significant deviations at both tails, which reflects the presence of outliers and a slightly heavier-tailed distribution compared to the male group.

Hypothesis Testing for Mean Body Temperatures by gender

Population variances are unknown and n,m are large. We check for difference between mean body temperature using Two Sample t-test for unequal sample variance also called Welch's t-test.

H0 : Mean_Body_Temperature(Male) = Mean_Body_Temperature(Female)
Ha : Mean_Body_Temperature(Male) != Mean_Body_Temperature(Female)

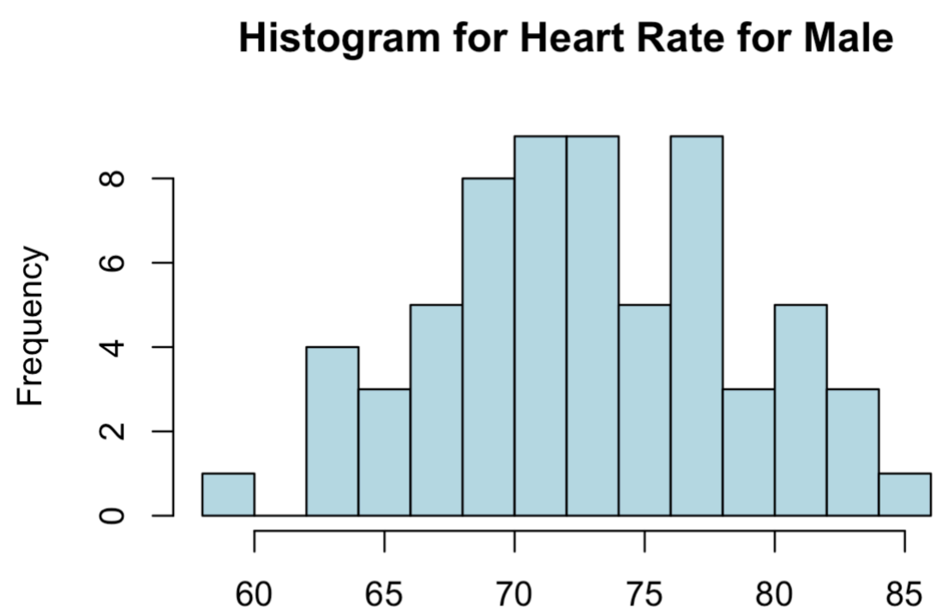
Assumption : Population variances are unknown. Significance Level = 5%
From the given sample we can calculate sample variances.

Results of t-test :
p-value = 0.02394 < 0.05 (=alpha)
p-value < alpha

At alpha = 0.05, we reject the null hypothesis.

We have enough evidence to conclude that on average, the body temperatures of the two genders are not the same.

b. **Exploratory Data Analysis for Heart Rate by Gender**

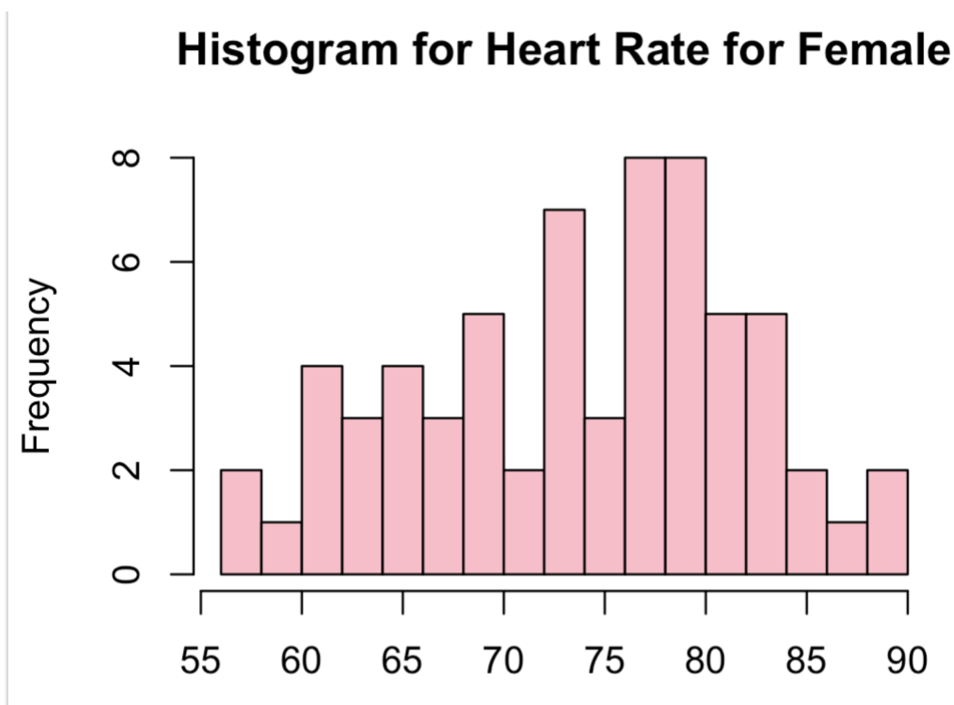


- Heart rate data for males appears to be approximately normal and symmetric, centered around a median of 73.00 and a mean of 73.37 beats per minute.

- Summary and Statistics for Male Heart Rate:

```
> summary(filter(heartrate, gender == 1)$heart_rate)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  58.00   70.00   73.00   73.37   78.00   86.00 
> sd(filter(heartrate, gender == 1)$heart_rate)
[1] 5.875184
> range(filter(heartrate, gender == 1)$heart_rate)
[1] 58 86
```

Histogram for Female Heart Rate :

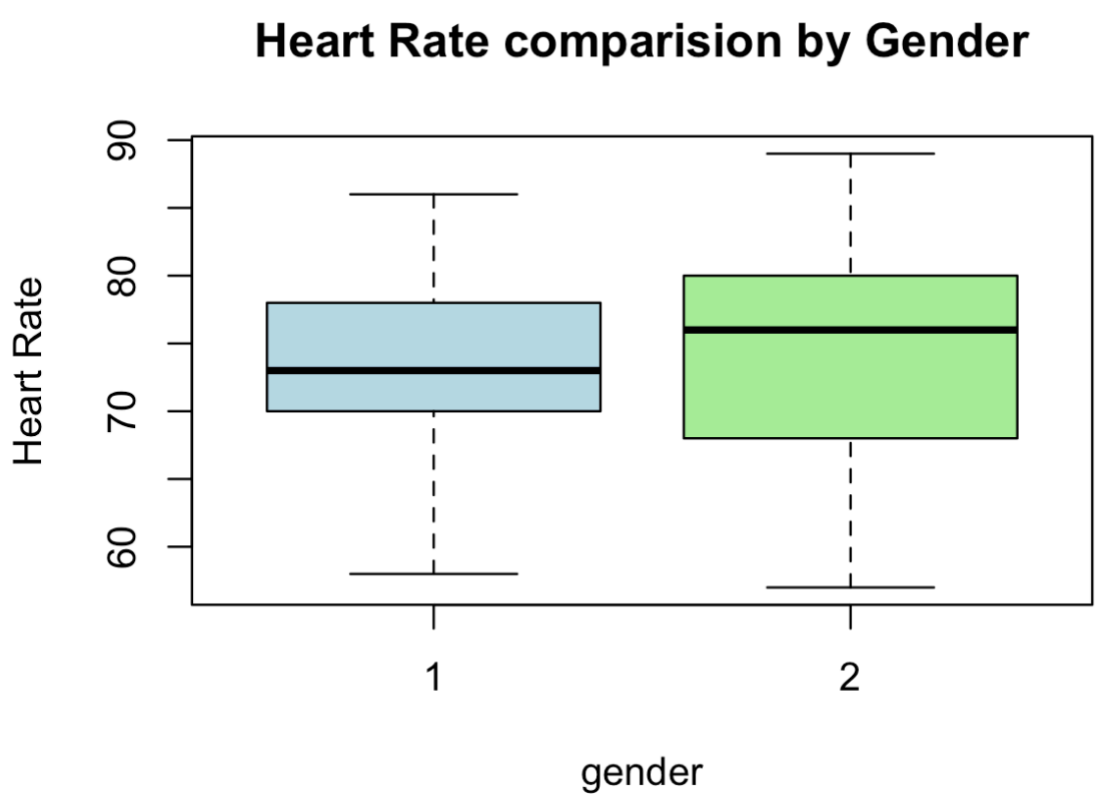


- Heart rate data for females appears to be roughly normal but slightly more spread out, centered around a median of 76.00 and a mean of 74.15 beats per minute.

Summary and Statistics for Female Heart Rate:

```
> summary(filter(heartrate, gender == 2)$heart_rate)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  57.00   68.00   76.00   74.15   80.00   89.00 
> sd(filter(heartrate, gender == 2)$heart_rate)
[1] 8.105227
> range(filter(heartrate, gender == 2)$heart_rate)
[1] 57 89
```

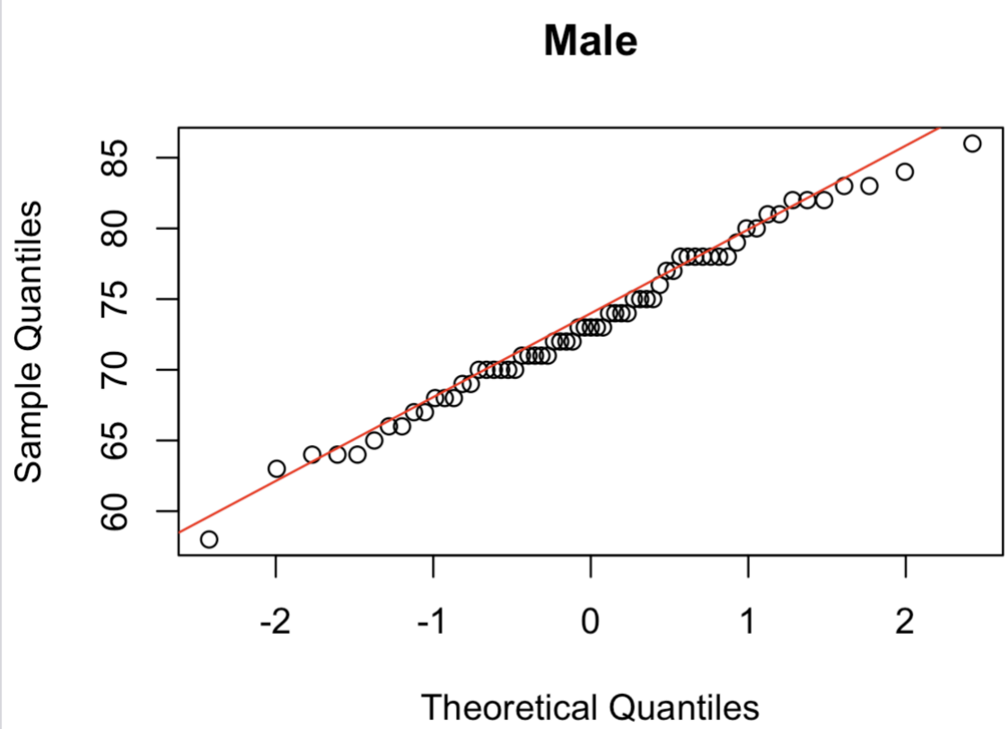
Side by Side Box Plot for Heart Rate by Gender:



- The side-by-side boxplot shows that the distribution of heart rate readings differs between the two genders. Females have a higher median heart rate compared to Male.

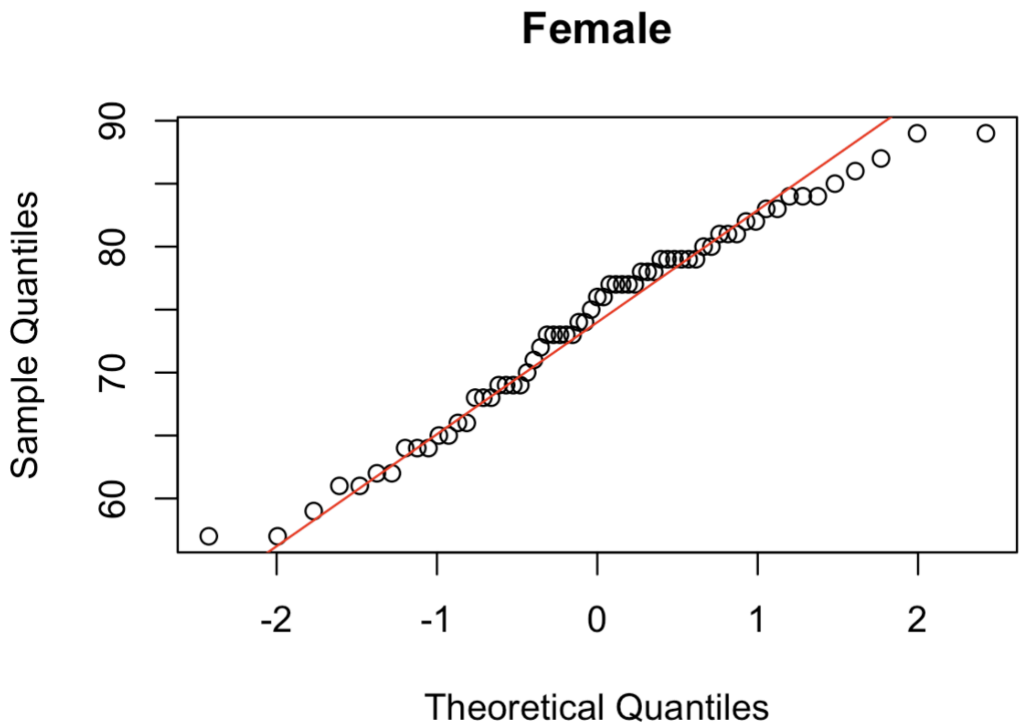
- The female group exhibits a larger interquartile range and a wider overall range than the male group, suggesting greater variability in heart rates among the females sampled.

Plot for Male Heart Rate :



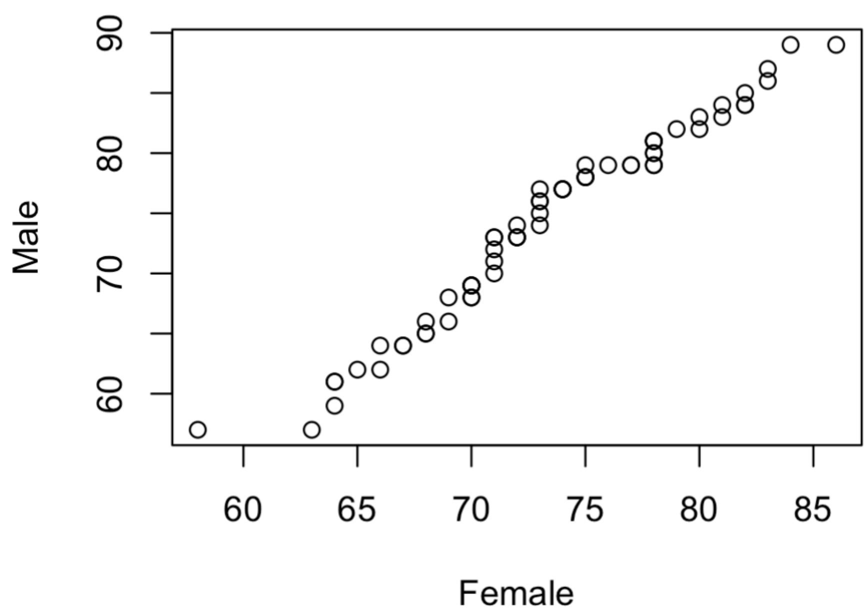
- The points in the male plot closely follow the straight red reference line across most of the range, indicating that the heart rate distribution for males is approximately normal.

Plot for Female Heart Rate :



- The female plot shows points following the linear reference line, suggesting a normal distribution. But, there is a slight deviation at the upper tail, indicating the distribution is slightly light-tailed compared to a perfect normal distribution.

QQ Plot of Male vs Female Heart Rate :



- The QQ Plot shows a strong positive linear relationship between the two groups. A few outliers visible at the lower end for both genders. This alignment indicates that as heart rates increase in one group, they tend to increase proportionally in the other group too.

Hypothesis Testing for Mean Heart Rates by Gender

Population variances are unknown and sample sizes are large. We check for difference between mean heart rate using **Two Sample t-test** for **unequal sample variance** also called **Welch's t-test.**

H0 : Mean_Heart_Rate(Male) = Mean_Heart_Rate(Female)
Ha : Mean_Heart_Rate(Male) != Mean_Heart_Rate(Female)

Assumption : Population variances are unknown. Significance Level = 5%
From the given sample we can calculate sample variances.

```
> t_result <- t.test(male_hr , female_hr,
+                    alternative = "two.sided",
+                    conf.level = 0.95,
+                    var.equal = FALSE)
> print(t_result)

        Welch Two Sample t-test

data:  male_hr and female_hr
t = -0.63191, df = 116.7, p-value = 0.5287
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -3.243732  1.674501
sample estimates:
mean of x mean of y
 73.36923  74.15385
```

Results of t-test :
p-value = 0.5287 > 0.05 (=alpha)
p-value > alpha

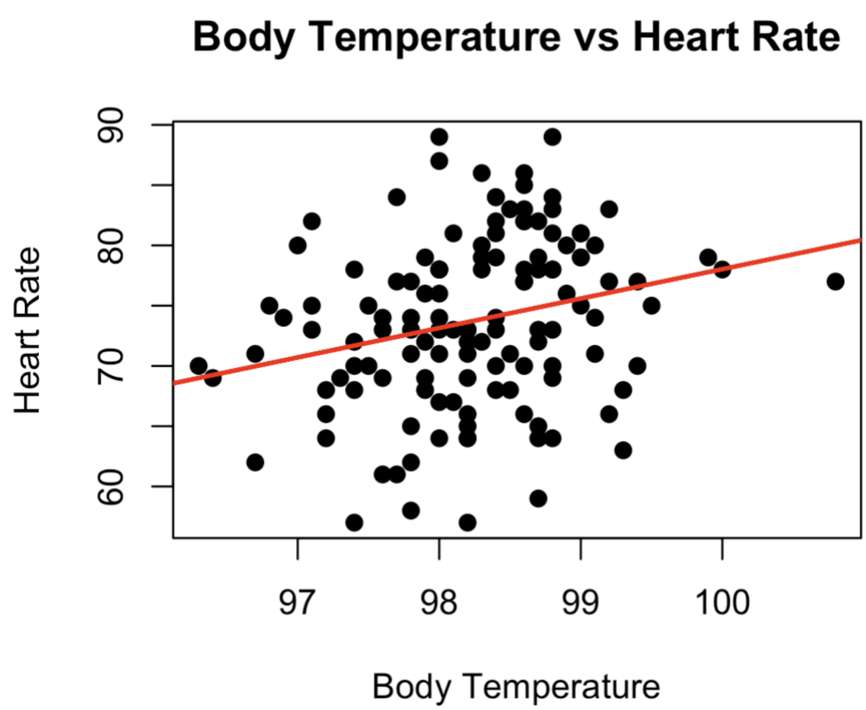
At alpha = 0.05, we fail to reject the null hypothesis.

Therefore, there is insufficient evidence to conclude that mean heart rate differs between males and females.

c. **Relationship between Heart Rate and Body Temperature**

- The scatter plot shows a positive upward trend, indicated by the red regression line. As body temperature increases, heart rate generally tends to increase as well.

Scatter plot with a regression line (next page)



H0 : No linear relationship between body temperature and heart rate
Ha : There is a linear relationship between body temperature and heart rate

Pearson correlation test

```
> # Correlation test
> cor.test(heartrate$body_temperature, heartrate$heart_rate)

	Pearson's product-moment correlation

data:  heartrate$body_temperature and heartrate$heart_rate
t = 2.9668, df = 128, p-value = 0.003591
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.08519113 0.40802170
sample estimates:
      cor 
0.2536564 
```

- The correlation test gives a p-value of 0.003591, which is significantly less than the alpha of 0.05. This allows us to reject the null hypothesis and conclude that a statistically significant linear relationship exists.
- The correlation coefficient (r) is approximately 0.25. This indicates a weak to moderate positive linear relationship. While the relationship is statistically "real," the high amount of scatter around the line suggests that body temperature only explains a small portion of the variation in heart rate.

QUESTION 2

To predict the PSA level for a representative patient, we set the quantitative predictors to their sample means and the qualitative predictor vesinv to its mode.

Model Summary :

```
> summary(model)

Call:                                          63.8659793814433
lm(formula = psa ~ cancervol + weight + age + benpros + vesinv +
    capspen + gleason, data = prostate_data)

Residuals:
    Min      1Q  Median      3Q     Max
-61.330  -8.130  -0.014   6.324 167.436

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -15.24264   40.53932  -0.376 0.707814
cancervol     2.03225    0.59359   3.424 0.000936 ***
weight        0.01132    0.07395   0.153 0.878708
age          -0.53721    0.47588  -1.129 0.261977
benpros       1.29831    1.20168   1.080 0.282878
vesinv1      19.60957   10.89184   1.800 0.075187 .
capspen       1.09877    1.33377   0.824 0.412253
gleason       7.05922    5.19452   1.359 0.177589
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 31.17 on 89 degrees of freedom
Multiple R-squared:  0.4585,	Adjusted R-squared:  0.4159
F-statistic: 10.77 on 7 and 89 DF,  p-value: 9.266e-10
```

From model summary, we get the regression equation as :

PSA = -15.24 + 2.03(cancervol) + 0.01(weight) - 0.54(age) + 1.30(benpros) + 1
9.61(vesinv1) + 1.10(capspen) + 7.06(gleason)

- cancervol is the most significant predictor with $p < 0.001$. For every 1 cc increase in cancer volume, the PSA level is expected to increase by approximately 2.03 mg/ml, holding all other variables constant.
- The coefficient for vesinv1 = 19.61. This indicates that, on average, patients with seminal vesicle invasion have PSA levels 19.61 units higher than those without it, though this is only marginally significant at the 0.10 level.
- Adjusted R-squared = 0.4159 : The model explains about 41.6% of the variance in PSA levels after adjusting for the number of predictors.

- F-statistic: 10.77 with a p-value of 9.266e-10, indicating that the model as a whole is statistically significant.

Result: The predicted PSA level for this patient is 19.48 mg/ml.

SECTION 2

QUESTION 1

A. Body Temperature Comparison

Exploratory Data Analysis Code

```r
library(pacman)

pacman::p_load(rio)

heartrate <- import("/Users/fathimafaridamohammad/Downloads/bodytemp-heartrate.csv")

head(heartrate)

hist(heartrate$body_temperature [heartrate$gender == "1"],
     breaks = 15, main = "Histogram for Body Temperature for Male",
     xlab = "", col="lightblue")

hist(heartrate$body_temperature [heartrate$gender == "2"],
     breaks = 15, main = "Histogram for Body Temperature for Female",
     xlab = "", col="pink")

pacman::p_load(dplyr)

boxplot(`body_temperature` ~ gender , data = heartrate,
        col = c("lightblue" , "lightgreen"),
        main = "Body Temperature comparision by Gender", ylab = "Body Temperature")
```

Summary Statistics for Male and Female Body Temperature

```r
## Summary statistics for Male Body Temperature
summary(filter(heartrate, gender == 1)$body_temperature)
sd(filter(heartrate, gender == 1)$body_temperature)
range(filter(heartrate, gender == 1)$body_temperature)
## Summary statistics for Female Body Temperature
summary(filter(heartrate, gender == 2)$body_temperature)
sd(filter(heartrate, gender == 2)$body_temperature)
range(filter(heartrate, gender == 2)$body_temperature)
## QQ Line for Male Body Temperature
qqnorm(heartrate$body_temperature[heartrate$gender == 1], main = "Male")
qqline(heartrate$body_temperature[heartrate$gender == 1] , col = "red")
## QQ line for Female Body Temperature
qqnorm(heartrate$body_temperature[heartrate$gender == 2], main = "Female")
qqline(heartrate$body_temperature[heartrate$gender == 2], col = "red")
```

Hypothesis Testing for Mean Body Temperatures

```r
male_bt <- filter(heartrate, gender == 1)$body_temperature

female_bt <- filter(heartrate, gender == 2)$body_temperature
## Two Sample T Test or Welsh Test
t_result <- t.test(male_bt , female_bt,
                   alternative = "two.sided",
                   conf.level = 0.95,
                   var.equal = FALSE)

print(t_result)
```

B.Heart Rate Comparison

Exploratory Data Analysis Code

```r
library(pacman)

pacman::p_load(rio)

heartrate <- import("/Users/fathimafaridamohammad/Downloads/bodytemp-heartrate.csv")

head(heartrate)

hist(heartrate$heart_rate [heartrate$gender == "1"],
     breaks = 15, main = "Histogram for Heart Rate for Male",
     xlab = "", col="lightblue")

hist(heartrate$heart_rate [heartrate$gender == "2"],
     breaks = 15, main = "Histogram for Heart Rate for Female",
     xlab = "", col="pink")

pacman::p_load(dplyr)

boxplot(`heart_rate` ~ gender , data = heartrate,
        col = c("lightblue" , "lightgreen"),
        main = "Heart Rate comparision by Gender", ylab = "Heart Rate")
```

Summary Statistics for Male and Female Heart Rate

```r
## Summary statistics for Male
summary(filter(heartrate, gender == 1)$heart_rate)
sd(filter(heartrate, gender == 1)$heart_rate)
range(filter(heartrate, gender == 1)$heart_rate)
## Summary statistics for Female
summary(filter(heartrate, gender == 2)$heart_rate)
sd(filter(heartrate, gender == 2)$heart_rate)
range(filter(heartrate, gender == 2)$heart_rate)
## QQ Line for Male Heart Rate
qqnorm(heartrate$heart_rate[heartrate$gender == 1], main = "Male")
qqline(heartrate$heart_rate[heartrate$gender == 1] , col = "red")
## QQ line for Female Heart Rate
qqnorm(heartrate$heart_rate[heartrate$gender == 2], main = "Female")
qqline(heartrate$heart_rate[heartrate$gender == 2], col = "red")
```

Hypothesis Testing for Mean Heart Rates

```r
male_hr <- filter(heartrate, gender == 1)$heart_rate

female_hr <- filter(heartrate, gender == 2)$heart_rate
## Two Sample T Test or Welsh Test
t_result <- t.test(male_hr , female_hr,
                   alternative = "two.sided",
                   conf.level = 0.95,
                   var.equal = FALSE)

print(t_result)
```

C. Linear Relationship

Code to plot a scatter plot

```r
# Scatter plot
plot(heartrate$body_temperature,
     heartrate$heart_rate,
     main = "Body Temperature vs Heart Rate",
     xlab = "Body Temperature",
     ylab = "Heart Rate",
     pch = 19)

# Add regression line
abline(lm(heart_rate ~ body_temperature, data = heartrate),
       col = "red",
       lwd = 2)
```

Correlation Coefficient calculation

```r
# Correlation coefficient
cor(heartrate$body_temperature, heartrate$heart_rate)

# Correlation test
cor.test(heartrate$body_temperature, heartrate$heart_rate)
```

QUESTION 2

Loading dataset and model summary :

```r
# Load  dataset
prostate_data <- read.csv("/Users/fathimafaridamohammad/
                          Downloads/prostate_cancer.csv")

# To treat 'vesinv'as a qualitative variable
prostate_data$vesinv <- as.factor(prostate_data$vesinv)

# Building the linear model
model <- lm(psa ~ cancervol + weight + age + benpros +
              vesinv + capspen + gleason,
            data = prostate_data)

# Summary of the model to see coefficients
summary(model)

# Prepare data for prediction
# Calculate means for quantitative predictors
mean_cancervol <- mean(prostate_data$cancervol, na.rm = TRUE)
mean_weight    <- mean(prostate_data$weight, na.rm = TRUE)
mean_age       <- mean(prostate_data$age, na.rm = TRUE)
mean_benpros   <- mean(prostate_data$benpros, na.rm = TRUE)
mean_capspen   <- mean(prostate_data$capspen, na.rm = TRUE)
mean_gleason   <- mean(prostate_data$gleason, na.rm = TRUE)

# Find the most frequent category for 'vesinv'
mode_vesinv <- names(sort(table(prostate_data$vesinv),
                          decreasing = TRUE))[1]
```

Running the model on new patient :

```r
# Create a data frame for the new patient
new_patient <- data.frame(
  cancervol = mean_cancervol,
  weight    = mean_weight,
  age       = mean_age,
  benpros   = mean_benpros,
  vesinv    = as.factor(mode_vesinv),
  capspen   = mean_capspen,
  gleason   = mean_gleason
)

# Predict the PSA level
predicted_psa <- predict(model, newdata = new_patient)

# Output the result
cat("Predicted PSA level for the average patient:",
    predicted_psa, "\n")
```

Final PSA level :

```
> cat("Predicted PSA level for the average patient:", predicted_psa,
"\n")
Predicted PSA level for the average patient: 19.48476
> 
```

THE END